

Artificial Intelligence

Chapter 1: Introduction to Artificial Intelligence

What is AI ??

- At its core, AI is about teaching computers to think, learn and solve problems like humans do.
- Instead of programming a computer with exact instructions for every single situation, we design to recognize patterns, learn from experience, and make its own decisions.
- AI consists of design of intelligent agents, which is a program that perceives its environment and takes action that maximizes its chance of success.
- Artificial Intelligence is the part of computer science concerned with designing intelligent computer systems, that is, systems that exhibit the characteristics we associate with intelligence in human behavior.

Why AI in today's tech world ??

- Solving complex problems through rational agents
 - Rational agents are designed to act in the best possible way to achieve their goals.
 - This approach is powerful because it's applicable to nearly any problem, from playing chess to driving a car or diagnosing diseases.
- Power of Machine Learning
 - Instead of being explicitly programmed, systems can learn from vast amounts of data to recognize patterns, make predictions and adapt.
- Transforming industries with specific applications
 - Natural language processing: Enables machines to understand and generate human language. This powers everything from voice assistants and chatbots to translation services.
 - Computer Vision: Allows machines to interpret and understand visual information from the world, crucial for medical image analysis, robotics and many more.
- Making sense of Big Data
 - The world generates a huge amount of data every second. AI provides tools to analyze this data and extract valuable insights.
 - This allows for better data driven decision-making in fields like finance, healthcare, and so on.
- Increasing efficiency and augmenting human abilities
 - AI can automate dangerous, repetitive and tedious tasks, freeing up humans for more creative and strategic work.
 - In many areas, AI doesn't replace humans but augments their abilities, helping doctors make better diagnosis or scientists accelerate research.

Applications of AI

- Natural Language Processing
- Computer Vision
- Machine Learning
- Robotics
- Healthcare and Medicine
- Finance and Banking
- E-Commerce and Retail
- Generative AI
- Gaming and Entertainment
- Agriculture
- Education

Different definitions of AI by different writers

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.	

- Thought Processes vs Behavior
 - Does AI focus on the internal reasoning and thinking of a machine or its external actions and behavior ?
- Human Performance vs Ideal Rationality
 - Is the goal to replicate human intelligence or to achieve a perfect, logical rationality ?

Thinking Humanly: Cognitive Modeling approach

- This category defines AI as effort to make computers think in the way as humans do.
- The goal is to create machines with “minds, in literal sense”.
- This approach is linked to the field of cognitive science, which seeks to create theory of human mind.
- To determine if a program thinks like a human, its processes would need to be compared to the patterns of human thought, often studied through introspection and psychological experiments.

Acting Humanly: The Turing test approach

- This approach focuses on machine's ability to perform actions that are indistinguishable from those of human.
- Turing test(designed by Sir Alan Turing) was a test in which a computer passes the test if human interrogator cannot tell whether the responses are from a person or a computer.
- A computer would to possess following capability to pass this test:
 - Natural language processing
 - Knowledge representation to store what it knows or hears.
 - Automated Reasoning to use stored information to answer and draw conclusions.
 - Machine learning to adapt to new circumstances to detect and extrapolate patterns.

Thinking Rationally: The “law of thought” approach

- This category views AI as study of computations that make it possible to perceive, reason and act in a rational manner.
- It is based on generating correct inference and building systems based on formal logic.
- Aristotle was the first one who attempted to codify “right thinking”.
- For example: “Socrates is a man, All men are mortal: therefore, Socrates is mortal.”
- There are mainly two problems to this approach:
 - First it is not easy to take informal knowledge and represent in the formal terms.
 - Second there is big difference between solving a problem “in principle” and actually solving in practice.

Acting rationally: The rational agent approach

- This is the most prevalent approach in modern AI.
- It defines AI as the study and design of intelligent agents, which are the systems that perceive their environment and take actions to achieve the best possible outcome.
- This approach is more general than “thinking rationally” because making correct inferences is just one of the way to act rationally.
- A rational agent is not concerned with thinking like human, rather with achieving its goals in the most effective way possible.

Thank YOU

Introduction to Artificial Intelligence

Lecture 2

Foundations of AI

- Different fields have contributed to AI in the form of ideas, viewpoints and techniques.
- Philosophy:
 - Logic, reasoning, mind as a physical system, foundations of learning, language and rationality.
 - Where does knowledge come from ?
 - How does knowledge lead to action ?
 - How does mental mind arise from physical brain?
 - Can formal rules be used to draw valid conclusions?
- Mathematics
 - Formal representation and proof algorithms, computation, undecidability, intractability, probability
 - What are the formal rules to draw the valid conclusions ?
 - What can be computed ?
 - How do we reason with uncertain information ?

Foundations of AI

- Economics
 - Formal theory of rational decisions, game theory, operation research
 - How should we make decisions so as to maximize the payoff ?
 - How should we do this when others may not go along ?
 - How should we do this when the payoff may be far in the future ?
- Linguistics
 - Knowledge representation, grammar
 - How does language relate to thought ?
- Psychology
 - Adaptation, phenomena of perception and motor control
 - How humans and animals think and act ?
- Neuroscience
 - How do brains process information?
- Control theory
 - How can artifacts operate under their own control ?

Brief history of AI

- **The Gestation of Artificial Intelligence (1943–1955)**

- This period laid the intellectual groundwork for AI before it was a formal field.
- **1943:** Warren McCulloch and Walter Pitts model a simple **artificial neuron** using electrical circuits, showing that networks of them could compute any logical function.
- **1949:** Donald Hebb proposes the **Hebb rule** for how neurons learn by strengthening connections between them.
- **1950:** Alan Turing publishes "Computing Machinery and Intelligence," introducing the **Turing Test**, machine learning, and other core concepts.
- **1951:** Marvin Minsky and Dean Edmonds build **SNARC**, the first computer specifically designed to simulate a neural network.
- **1950s:** Early chess and checkers programs are written by researchers like Claude Shannon and Arthur Samuel, the latter of which could learn from its experience.

- **The Birth of Artificial Intelligence (1956)**

- This is the single event that officially launched the field.
- **1956:** The **Dartmouth Workshop**, a two-month summer workshop organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon.
- It was here that John McCarthy officially coined the term "**Artificial Intelligence**."
- The attendees—considered the founding fathers of AI—were incredibly optimistic, believing that a machine as intelligent as a human being was no more than a generation away.

Brief history of AI

- **Early enthusiasm, great expectations(1952-1969)**

- **1958:** John McCarthy invents the **Lisp** programming language, which becomes the dominant language for AI research.
- **1959:** Newell and Simon develop the **General Problem Solver (GPS)**, a program designed to imitate human problem-solving strategies.
- **1966:** The book "Machine Translation" reports on the failures of translation projects, a major early setback.
- **Late 1960s:** Minsky's students develop programs that can solve problems in limited domains like algebra word problems and geometric analogies.

- **A Dose of Reality (1966–1973)**

- Also known as “AI winter”.
- Early programs lacked scalability to deal with real world complexity.
- Many tasks require a huge amount of background or **common-sense knowledge**, which was difficult to program.
- Key theoretical limitations were discovered, such as Minsky and Papert's proof in their 1969 book *Perceptrons* that simple neural networks could not solve certain fundamental problems.
- These problems led to major funding cuts in AI researches.

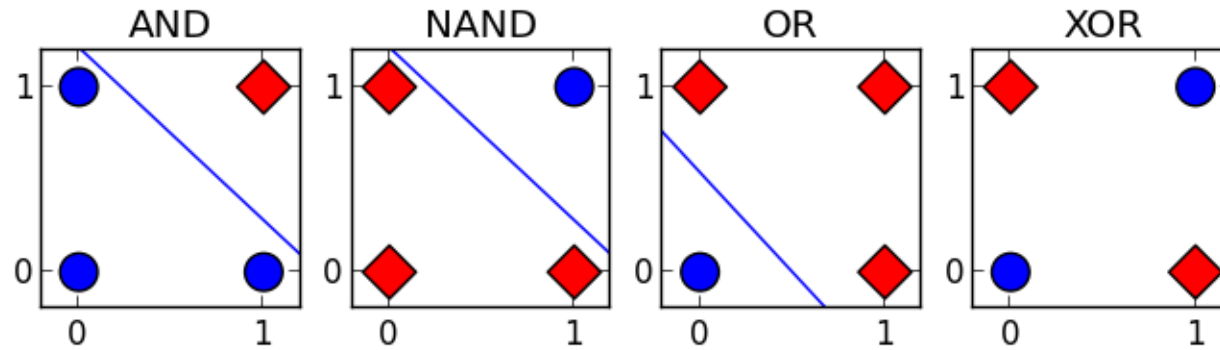
Brief history of AI

- **Knowledge based systems: The key to power?(1969-1979)**
 - The solution to the previous era's problems was to give programs more knowledge. This led to the rise of **expert systems**.
 - **DENDRAL (1969)**: An early expert system that could infer the molecular structure of organic compounds. Its power came from the vast amount of specialized knowledge it contained.
 - **MYCIN (mid-1970s)**: An expert system for diagnosing blood infections. It was notable for using **certainty factors** to represent a measure of belief in its own conclusions, a precursor to probability-based systems.
- **AI becomes an industry (1980 – present)**
 - Expert systems moved out of the lab and into the commercial world, leading to a massive investment boom.
 - **1982**: Digital Equipment Corporation (DEC) implements **R1**, an expert system that helped configure their computer systems, saving the company an estimated \$40 million a year.
 - **Mid-1980s**: A boom in AI companies selling specialized "Lisp machines" and expert system software.
 - **1988**: The expert system market collapses, leading to the **second "AI Winter."** The systems were expensive, difficult to update, and were out-competed by newer, cheaper hardware.
- **The Return of Neural Networks (1986 – present)**
 - While expert systems were fading, a different approach was making a comeback.
 - **1986**: The invention of the **back-propagation algorithm** is widely circulated. This allowed deep neural networks (with multiple layers) to be trained effectively, overcoming the earlier limitations identified by Minsky.

Brief history of AI

- **AI Adopts the Scientific Method (1987–present)**
 - The field moved away from building systems just to "see if they worked" and towards a more rigorous, mathematical, and data-driven approach.
 - **Judea Pearl's work on Bayesian networks** and probabilistic reasoning became highly influential.
 - The field began to rely on rigorous theorems and solid experimental evidence rather than just intuition.
- **The emergence of Intelligent Agents (1995 – present)**
 - AI shifted from solving isolated problems to unified goal of building “intelligent agents”. An agent is any system that perceives its environment with sensors and acts upon it with actuators to achieve a goal.
 - The concept of agents unified the entire field of AI (natural language processing, computer vision).
 - The objective of AI systems changed from mimicking human behavior to creating rational agents that could take best possible action to achieve its goals.
- **The availability of Very Large Data Sets (2001 – present)**
 - The rise of internet led to creation of massive datasets.
 - This data combined with powerful computers with high computation and deep learning algorithms fueled the modern AI revolution.

The XOR problem



- XOR problem showed the limitation of first-generation perceptron and were one of the major reason behind funding cut off in the field of AI then.
- When the data points are not linearly separable, it was impossible to create boundary for the first generation of perceptron.

THANK YOU !!!!!!!!!!!!!!!